SPLITTING A REGULAR SEQUENCE GENERATING J ALONG A SUBIDEAL I

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Injectivity on conormal level. lemma-injective (same chapter, named). Gives that I/mI -> J/mJ is injective, which splits the minimal generators.
- Regularity of minimal generating systems. more-algebra-lemma-independence-of-generators, more-algebra-lemma-noetherian-finite-all-equivalent (cross-chapter, named). Any minimal system of generators of J is a regular sequence; independence transfers to the initial segment and to the images in A/I.

Source reference. Original lemma 09PV: sources/stacks-dpa/pinned-original-source.tex, source lines 1790–1790. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 066A: focused-seeds/geometry/S9-dpa-regular-sequence/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09CC: focused-seeds/geometry/S9-dpa-regular-sequence/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a Noetherian local ring. Let $I \subset J \subset A$ be proper ideals. Assume*

(1) *A/J has finite tor dimension over A/I, and*
(2) *J is generated by a regular sequence.*

Then I is generated by a regular sequence and J/I is generated by a regular sequence.

Proof. By Lemma 09PV we see that $I/\mathfrak{m}I \to J/\mathfrak{m}J$ is injective. Thus we can find $s \leq r$ and a minimal system of generators $f_1, \ldots, f_r$ of J such that $f_1, \ldots, f_s$ are in I and form a minimal system of generators of I. The lemma follows as any minimal system of generators of J is a regular sequence by More on Algebra, Lemmas 066A and 09CC. □

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.